

Lesson 1

Characteristics of Vertebrates and Invertebrates



Do List

1. Describe the characteristics of vertebrates and invertebrates.
2. Give examples of vertebrates and invertebrates.
3. Identify ways by which animals adapt to their environment.

Energize Your Mind

The picture above shows a carabao. What kind of animal is it? Where do you think this animal lives? To which animal classification does carabao belong?

Equip Yourself

Animals are grouped into vertebrates and invertebrates. Vertebrates are animals with a backbone, whereas invertebrates are animals without a backbone. A backbone is the column of bones, muscles, and other tissues in the back of an animal. It extends from the bottom of the skull to the lower part of the trunk. It is also called the spinal column or the spine. The backbone serves as a pillar that supports the body's weight. It also protects the spiral cord.

Vertebrates

Vertebrates are divided into mammals, reptiles, amphibians, fish, and birds.

Mammals

Mammals are vertebrates that possess mammary glands, which are glands that produce milk for the young. In addition, most mammals are warm-blooded, have hair or fur, and give birth to live offspring. Examples of mammals include goats, bats, cats, tigers, whales, deer, and cows. Can you name other examples of mammals?

bat

- whale
- deer

- Sheep

Extend Your Understanding (Word Unlock)

A gland is an organ or a group of cells that produces substances that help the body work and grow.

Warm-blooded animals can keep their body temperature constant regardless of the temperature of their surroundings.

A mammal's life cycle has three stages, namely, before birth, while the animal is young, and as an adult. Before birth, most mammals undergo a period of development as an embryo or fetus inside the mother. This period is called gestation. One group of mammals, the marsupials, have a short gestation period and newborns come out undeveloped.

The babies then continue their development in a pouch-like sac called the other group's longer gestation period, and mothers give birth to completely developed young. a

In the second stage, young mammals are cared for by their mothers and nursed with milk.

During this period, the animals learn everything from their parents.

In the third stage, the young mammals become adults. At this stage, they look very much like their parents. It is also at this stage that they become ready to reproduce.

Extend Your Understanding (Know Zone)

Did you know that the whale is a mammal? Whales, along with dolphins, seals, and dugongs, are only some of the aquatic mammals that can be found on our planet.

Reptiles

- Reptiles are cold-blooded vertebrates that have skin covered with scales or bony plates, which prevent the skin from drying out. Most reptiles lay eggs. Examples of reptiles are snakes, turtles, lizards, alligators, and crocodiles.
- snake
- alligator
- turtle V

Most reptiles reproduce sexually. Reptile eggs are surrounded by an eggshell, which protects the eggs and allows them to be easily transported. The reptile embryo develops inside the egg until it is ready to hatch. Once it hatches from the egg, it becomes a hatchling that can fend for and defend itself. It then continues to grow until it becomes a juvenile, which looks similar to the adults but smaller. It becomes an adult once it becomes ready to reproduce.

Extend Your Understanding (Know Zone)

Cold-blooded animals have body temperatures that depend on the temperature of their surroundings. These animals cannot regulate their body temperature on their own. Instead, they rely on their environment to keep them warm or to lower their body temperature

Amphibians

Amphibians are cold-blooded vertebrates that inhabit a wide variety of ecosystems. Most live part of their lives in water and part on land. Thus, most amphibians undergo a process of great physical change and development called metamorphosis. They hatch from eggs and become larvae, which are inactive, underdeveloped young that are significantly different from the adults. At this stage, baby amphibians have external gills that allow their development in an aquatic environment. They continue their metamorphosis until they become adults. By this stage, they have developed lungs and limbs that allow them to live on land. Examples of amphibians are frogs, toads, salamanders, and newts.

- frog
- toad
- newt

Amphibians are good indicators of the health of an ecosystem. Because they breathe through their skin, they are highly sensitive to small changes in the environment. The presence of many amphibians in a habitat means that the ecosystem in that area is healthy.

Birds

Birds are vertebrates that have feathers as body covering and beaks for getting food. Some birds have strong muscular legs and claws for capturing their prey. These animals can regulate their body temperature, and they also lay eggs. Not all birds can fly. Some birds, such as penguins and kiwis, lost their ability to fly as a result of adaptation to their habitat.

- *Philippine eagle*
- *parrot*
- *Albatross*

Birds give birth to their young by laying eggs. Eggs are usually laid in nests, and kept warm by the parent's body heat until they hatch. Parental care continues after hatching, with the young provided with food and protection.

Some birds migrate from one area to another to maximize available breeding grounds and optimize food sources.

Examples of birds are the Philippine eagles, parrots, albatrosses, owls, and cockatoo

Fish

- *swordfish*
- *Saimon*

Fish are vertebrates that have fins. They are aquatic animals, which means they live in water. They have gills that allow them to breathe in water. Many fish have scales, but some do not. They are cold-blooded animals because they cannot regulate their own body temperature. For most fishes, egg development happens outside the body of the mother. Some retain their eggs inside them. Fishes in oceans can produce a large number of eggs.

Fishes are precious economic resources worldwide and are found in most bodies of water. Freshwater fishes include salmon and tawilis. Examples of saltwater fishes are tuna, sharks, and swordfish. Kyhe

Invertebrates Hailey

Invertebrates are animals without backbones. These animals include sponges, enidarians, flatworms, roundworms, segmented worms, echinoderms, mollusks, and arthropods.

Poriferans

- sea sponge
- sea sponge
- red sea sponge
- tube sponge

Poriferans are invertebrates that have pores. They are also called sponges. Most sponges attach themselves to rocks, plants, or other animals in an aquatic habitat. These animals are filter feeders because they consume fine food particles through their pores. These animals have irregular shapes, mostly that of a tube or has an opening at the top.

Sponges can produce asexually or sexually. Some sponges discharge eggs that are then fertilized in the water, but most keep their eggs until they hatch. When the eggs hatch, larvae break out and swim for a few days until they find a place to settle. They eventually become adult sponges.

Cnidarians

- hydra
- jellyfish
- sea anemone
- coral Vit

Cnidarians are invertebrates that are characterized by stinging cells called enidoblasts, which help in defense and in trapping food. Some cnidarians have tentacles. The body of cnidarians may be a hollow, vase-like structure called a polyp

or an umbrella-shaped structure called a medusa. Some midterins are always polyps, others are always mecusa, and some are a polyp in one stage of life and a medusa in another. After fertilization, the egg develops into a larva. The larva eventually swims to find a good location and becomes a polyp. It grows until it changes into the medusa form. The body then develops until the endian becomes an adult.

Cnidarians are mostly found in marine habitats, only a few species may be found in freshwater environments. Hydras, jellyfish, sea anemones, and corals belong to this group.

Flatworms

Flatworms are invertebrates that have a flat body with bilateral symmetry. This means that the right and left halves of their body are the same. The head of a flatworm functions as a region for sensing light, chemicals, and pressure. These animals move by contracting the muscle cells that lie below the surface of their skin. Examples of flatworms are planarians, tapeworms, and flukes. These animals can be found in saltwater, freshwater, or moist land habitats. Except for the

planarians, all flatworms are parasitic. This means that they live in or on other organisms (the hosts) and get their nutrients from or at the expense of the hosts.

hammerhead flatworm

5

tapeworm

fluke

Through asexual reproduction, flatworms divide periodically, thus creating new worms. In sexual reproduction, some perform a mating ritual that eventually results to several hundred eggs. Eventually, free swimming larvae or small worms come out. Other flatworm life cycles are more complex. The eggs of parasitic flatworms are digested by host animals. Within the hosts, the eggs develop and the worms come out as larvae which further infect the host. The hosts may be vertebrates or invertebrates.

Roundworms/Nematodes

Nematodes or roundworms are invertebrates with a cylindrical body. Like tapeworms, roundworms have bilateral symmetry. They can be free-living (not parasitic) or parasitic and sometimes cause diseases in humans, plants, and animals. Examples of roundworms are hookworms, ascaris, and pinworms. These worms may

live in freshwater, seawater, or terrestrial habitats. Like flatworms, roundworms can reproduce asexually through division. However, during sexual reproduction, roundworm eggs are fertilized and then hatched to release larvae. For parasitic roundworms, the animal host can ingest eggs from the environment. These eggs develop inside the host's body and become adult roundworms. The whole cycle is repeated after a number of weeks. Roundworms are the most common human parasites. It is important to wash your hands before eating because dirty hands infected with roundworm eggs are the usual source of these parasitic infections.

ascaris

8/28

7

8

10

165

Segmented Worms/Annelids

hookworm

Trichinella spiralis

Annelids or segmented worms are invertebrates that have a body with segments or rings that are joined together. Like flatworms and roundworms, these animals display bilateral symmetry. Examples of segmented worms are earthworms, leeches, and rag worms. Segmented worms can be found in freshwater, marine, and terrestrial habitats.

69%

1

ragworm

leech

earthworm

Some annelids reproduce sexually, others reproduce asexually, and some do both. If reproduction is done sexually, some annelids mate only once, whereas others mate continuously. Eggs are fertilized, and larvae are eventually released. These larvae later grow into adults. Annelids perform asexual reproduction by dividing themselves into two or more pieces.

Echinoderms

Echinoderms are invertebrates that can be recognized by the spiny skin that covers most of their bodies. Their body has radial symmetry, which means that they have similar parts that are arranged around a central axis. Echinoderms also have an internal skeleton or endoskeleton, and they use a water-based pressure system for breathing, eating, and moving. Examples of animals in this group are starfishes, sea urchins, and sea cucumbers.

sea urchin

VII

sea cucumber

starfish

Echinoderms can reproduce both sexually and asexually. In sexual reproduction, most echinoderm eggs are fertilized in open water. In some, the eggs are carried in special pouches where the embryos are given nutrients while they are developing. The larvae are eventually released and become adults, which reach sexual maturity in three years. In asexual reproduction, echinoderms split or divide themselves into two or more parts in a process called fragmentation. Each individual piece then grows the missing body parts, and a new animal grows.

Mollusks

Mollusks are invertebrates that have a soft, unsegmented body with bilateral symmetry. Many mollusks have one or two shells, which may be internal (inside) or external (outside). Mollusks may have a muscular foot, a set of tentacles, or both. Examples of mollusks are slugs, snails, clams, squid, mussels, scallops, oysters, and octopuses.

slug

octopus

giant clam

snail

0

2

Mollusks are highly diverse and make up about a fifth of all organisms that live in the oceans. They have a rough, tongue-like structure called a radula that is covered with numerous tiny teeth. The radula is used to scrape food particles off of rocks or to tear food into smaller pieces. Most mollusks have sensors for detecting substances, touch, and vibrations. Some also have eyes. Mollusks reproduce sexually and mainly through external fertilization. This means that the eggs are released and fertilized outside the body. However, some mollusks fertilize their eggs internally. All mollusks produce eggs, which eventually hatch into larvae that develop into adults.

Arthropods

Arthropods are invertebrates with exoskeletons, jointed appendages, and segmented bodies consisting of the head, thorax, and abdomen. Insects, spiders, millipedes, centipedes, crabs, lobsters, and shrimps are all arthropods. Arthropods lay eggs. They have antennae as their sense organs. Arthropods that live in water have eggs that are fertilized either internally or externally. On the other hand, arthropods that live on land undergo internal fertilization only. Only scorpions have their eggs hatched inside them when the mother gives birth to live offspring. Once arthropod eggs are hatched, the offspring may vary from small versions of the adult or undergo metamorphosis that includes a larval stage, until it becomes a fully developed adult

crab

spider 10

Millipede

Extend Your Understanding (Word Unlock)

An exoskeleton is a skeleton that is found outside the body, on the other hand, an endoskeleton is a skeleton that is found inside the body.

An appendage is a body part that sticks out from the main body.

Examples of appendages are arms, legs, and antennae. A jointed appendage is an arm, leg, or other appendage that consists of separate sections that are joined together by a joint. Joints (such as those of the shoulders, elbows, and knees) allow greater movement For the body part.

Animals can be classified into vertebrates or invertebrates.

Vertebrates are animals with a backbone.

Invertebrates are animals without a backbone.

Vertebrates include mammals, reptiles, amphibians, fish, and birds.

Invertebrates include poriferans, cnidarians, flatworms, roundworms. segmented worms, echinoderms, mollusks, and arthropods.

You may already know some of the invertebrate animals mentioned in the lesson. However, you may still be unfamiliar with the others. Learn more about the characteristics of some invertebrates by doing the activity that follows.

Explore Around

DESIGNING MY INVERTEBRATES

What You Need

pencil

sheets of bond paper
scissors
sheets of construction paper
Glue

What to Do

1. Choose three invertebrates. You may choose animals that you already know or those that you are not yet familiar with.
2. On separate sheets of bond paper, draw pictures of your selected invertebrates.
3. Design and decorate your drawings using the sheets of construction paper.
4. On a separate sheet of bond paper, write information about the invertebrates that you have selected.
5. Share your output with the class.

What You Have Learned

Answer the following questions. Write your answers in the space provided.

1. In your opinion, which is the most interesting among your chosen invertebrates? Why?
2. How would you compare and contrast your selected invertebrates?

How You Learned

1. Did you learn something new from the activity? What is it? Explain briefly.
2. What skills did you apply in performing the activity?

Express Your Learning

Answer the following questions:

1. What is the difference between the physical characteristics of vertebrates and invertebrates?
2. How are mammals different from other groups of animals?
3. How do you feel about the lesson? Explain your answer.

Lesson 2

Economic Importance of Vertebrates and Invertebrates

Do List

Describe the economic importance of vertebrates and invertebrates.

Energize Your Mind



Do you eat fish, clams, crabs, and chicken? Do you know what kind of animals they are? These animals provide you with foods that are rich sources of protein. There are plenty of animals that can give you nutrients. Animals can also be sources of raw materials for creating things. Can you think of other uses of animals?

Equip Yourself

Economic Value of Vertebrates

Fish are important sources of proteins. The different species of fish are not only sources of food but also of different products, such as fish oils, pet food, and fertilizers.

Birds such as chicken, duck, and quail are sources of eggs and meat. Birds also help control pests because they eat insects that can damage plants. They help in plant reproduction because they can transfer pollen from one plant to another or pick seeds and drop them to the ground to grow. *Amphibians* such as frogs are very useful in scientific studies, mainly because of their ability to undergo metamorphosis. *Frogs* are also used to control pests in farms.

Snakes, turtles, and crocodiles are reptiles that may also help control pests because they eat other animals that can damage plants and certain habitats. Lizards also eat insects found in your house. The venom of snakes, in particular, is used in preparing antivenom, or medications used to treat poisonous snake bites. Some reptiles are consumed as food, and some are used in research as samples or subjects.

Mammals, such as horse, carabao, camel, and elephant, are used to transport people and goods. Other mammals are sources of dairy products, meat, leather, and wool

Economic Value of Invertebrates

Like the vertebrates, invertebrates also have economic importance. Some invertebrates serve as sources of food for human and animal consumption. Invertebrates, like insects, help in plant reproduction because they act as pollinators. They also help maintain the balance of nature as some invertebrates can decompose matter and recycle nutrients on Earth by participating in a biochemical cycle.. Certain invertebrates can also be utilized in industries and research. In some

Seashells can be sources of food and nutrients.

Earthworms help decompose matter and improve the fertility of soil.

Vertebrates can be used as sources of food, raw materials, and antivenom. They are also used for medical and environmental research. for pest control, and for balancing nature.

Invertebrates can be used as sources of food, agents of nutrient cycle, and pollinators. They also maintain the balance of nature.

Share with your classmates the information you have learned about the economic value of vertebrate and invertebrate animals using your output in the activity that follows. Cherka

Explore Around

ANIMAL PAMPHLET

What You Need

book, magazine, or computer with internet connection

markers or colored pens

sheets of colored paper

scissors

Glue

What to Do

1. Get pictures of two vertebrates and one invertebrate from an old magazine or the internet
2. Do a research on the economic importance of the animals you have selected by gathering information about them from books, magazines, or the internet.
3. Make a pamphlet about the animals. Do the following steps:
 - a. Get a sheet of colored paper and fold them crosswise into three sections, as shown on the right.
 - b. On the front cover, write the title of your pamphlet and other information that may be required by your teacher such as your name, grade, and section.
 - c. On the first inside section of your pamphlet, paste a picture of one of your chosen animals and write the information that you have gathered about the importance of this animal.
 - d. Do the same on the succeeding sections for the other two animals.
 - e. Design your pamphlet and highlight important information about the animals.
4. Have your classmates see and read the contents of your output.

What You Have Learned

Answer the following questions. Write your answers in the space provided.

1. What selected?
2. What do the highlighted parts tell about the invertebrates that you have selected?

How You Learned

1. What new learning did you get from this activity?
2. What concerns did you encounter while making the pamphlet?

Express Your Learning

Write a two-paragraph essay about the economic importance of animals.

Lesson 3

Animals Endemic to the Philippines



Do List

1. Name some animals that are endemic to the Philippines.
2. Describe some animals that are endemic to the Philippines.

Energize Your Mind

Look at the pictures above. Are the animals in the picture familiar to you? Have you seen these animals? These animals are endemic to the Philippines.

Equip Yourself

The Philippines is a country rich in wildlife. Although many of the animals in the Philippines can be found in other countries, there are animal species that can only be found in the Philippines. Animals that are commonly found and present in a specific area are called endemic animals. The animals endemic to the Philippines have unique characteristics. What makes them unique? Find out as you study this lesson.

Photograph by Lin Yan is licensed under CC BY-SA 3.0

Philippine Eagle

The Philippine eagle, the national bird of the Philippines and one of the world's rarest birds of prey, is also called the monkey-eating eagle because it is believed to have been preying on monkeys. Studies show that it also eats animals such as large birds, snakes, and lizards. It inhabits the forests and is called by its local names agila, haribon, and banog.

The length of the body of the Philippine eagle is between 86 and 100 cm. It has a wingspan of about 2 m. and weighs between 5 and 8 kg. It has brown feathers and white belly and

underwing. The brown feathers on their head and nape form a crest. This bird of prey has strong legs, powerful claws, and a large bill.

Philippine eagles are threatened by hunting and by the destruction of their habitats. Some eagles are kept safe in captivity where attempts are made to breed them.

Tamaraw

The rare tamaraw, also called the Mindoro dwarf buffalo, is endemic to the island of Mindoro. This mammal is shorter and stockier than the carabao. It is hairy, with a dark brown to grayish color. It has a heavyset body, four legs that end in hooves, a short neck, and a small head with horns that are V-shaped, which have outer surfaces that are worn out and inner sides that are rough. Tamaraws, which prefer to be solitary, live in areas such as marshes and grasslands, near sources of water and food such as grasses, bamboo shoots, and wild sugarcane. Tamaraw population has declined due to human activities such as hunting and building of infrastructure in its habitat.

Philippine Crocodile

The Philippine crocodile, known locally as buwaya, is a small, freshwater crocodile. It has a broad snout and thick bony plates on its back. It has a golden-brown color which darkens as it reaches adulthood. This crocodile, which eats fish, is generally viewed as a human-eater. In reality, it will not attack people unless threatened or provoked.

Philippine Tarsier

The Philippine tarsier is found in the Visayan islands of Bohol, Samar, and Leyte, and some parts of Mindanao. It derives its name from its elongated tarsus or ankle bone, which allows it to jump about 3 m from tree to tree. It is between 85 to 160 mm in height and usually weighs 160 g or less. The average size of an adult tarsier is about the size of an adult human fist.

One of the features of the Philippine tarsier is its pair of eyes that are huge in proportion to its head. The eyes are attached to the skull, thus cannot move in its sockets. The neck is specially adapted to allow its head to rotate 180°. The color of its thin and rough fur ranges from gray to dark brown. Its tail, bald with a bunch of hair at the end and usually twice as long as its body length, allows the tarsier to keep its balance. The Philippine tarsier is an animal that is shy and keeps itself hidden from humans and predators. It sleeps during the day in tree trunks, shrubs, and bushes, and becomes very active at night. With its keen sight, it moves around trees and feeds on insects, spiders, and other small animals.

Tawilis

The freshwater fish *Sardinella tawilis*, locally known as tawidis, is found exclusively in Taal lake. It is a small fish that can be 15 cm long and can weigh 30 g or less. Its body is compressed laterally and covered with tough scales. This fish roams the volcanic lake in schools and feeds on plankton on the water's surface. Fried tawilis is a popular fish food in the country. It is also processed into other products as dried fish or cooked and bottled in oil.

tod Dee inc

Photograph by Gregg Tan is leased under CO HY-SASD

Some animals that are endemic to the Philippines are the Philippine eagle, tamaraw, Philippine crocodile, tarsier, and tawilis.

Make an illustration of the endemic animals you learned about in this lesson. Do the activity below,

Explore Around

ENDEMIC ANIMAL PICTURE

What You Need

magazine, newspaper, or computer with internet connection

scissors

illustration board

pencil

colored pens

Glue

What to Do

1. Look in magazines, newspapers, or the internet for pictures of at least three animals that are endemic to the Philippines.
2. Cut out or print the pictures.
3. Draw a natural animal habitat on the illustration board. It can be a forest or a body of water, depending on your chosen animals. Color your drawing.
4. Paste the pictures of animals on the illustration board.

What You Have learned

Answer the following questions based on the activity. Write your answers in the space provided.

1. How would you describe each animal that you have selected for the activity?
2. How would you differentiate these animals from one another?

How You Learned

1. What problem did you encounter in doing the activity? How did you resolve it?
2. Which part of the activity caught your interest? Why?

Express Your Learning

Give the distinguishing characteristics of the following rare animals.

1. Philippine eagle
2. tamaraw
3. Philippine crocodile

Lesson 4

Protecting and Caring for Animals



Do List

1. Describe some ways of protecting and caring for animals.
2. Practice ways of protecting and caring for animals.

Energize Your Mind

Do you have pets at home? How do you care for them? Animals, including your pets, do not have the capability to talk to people and express their needs. They need people to care for them and to provide their basic needs.

Equip Yourself

Ways of Protecting and Caring for Animals

Animals need your protection and care. You can do the following to protect and care for animals:

1. Preserve their natural habitat.
2. Provide enough space in which your pet animals can move around.
3. Support every effort and campaign to ban the capture, sales, transport, and kill-ing of exotic and endangered animals.
4. Join programs that: provide shelter to animals, especially to the endangered ones.

Remember that whatever their size or kind, animals are living things that need to be cared for and protected.

Protection and care for animals can be shown by preserving their habitats, not supporting activities that will endanger animals; and getting involved in environmental programs that protect animals, especially the endangered ones.

Conservation of wildlife is important. Without the active movement to protect animals, some will be hunted or disappear completely. There was a time when pandas were near extinction. Those who care for the environment and conservation fought to protect these animals. Now they are

no longer an endangered species! Learn more about the different ways to protect animals by doing the research activity below.

Explore Around

RESEARCH ON PROTECTING ANIMALS

What You Need

books, science magazines, or computer with internet connection
pen
sheets of bond paper

What to Do

1. Research on one conservation method for caring and protecting animals.
2. Your research should include the following information:
 - a. name of method
 - b. description of method
 - c. process of conservation
 - d. examples of animals that benefit from the method of conservation
3. Write your research findings on sheets of bond paper. Share your output with the class.

What You Have Learned

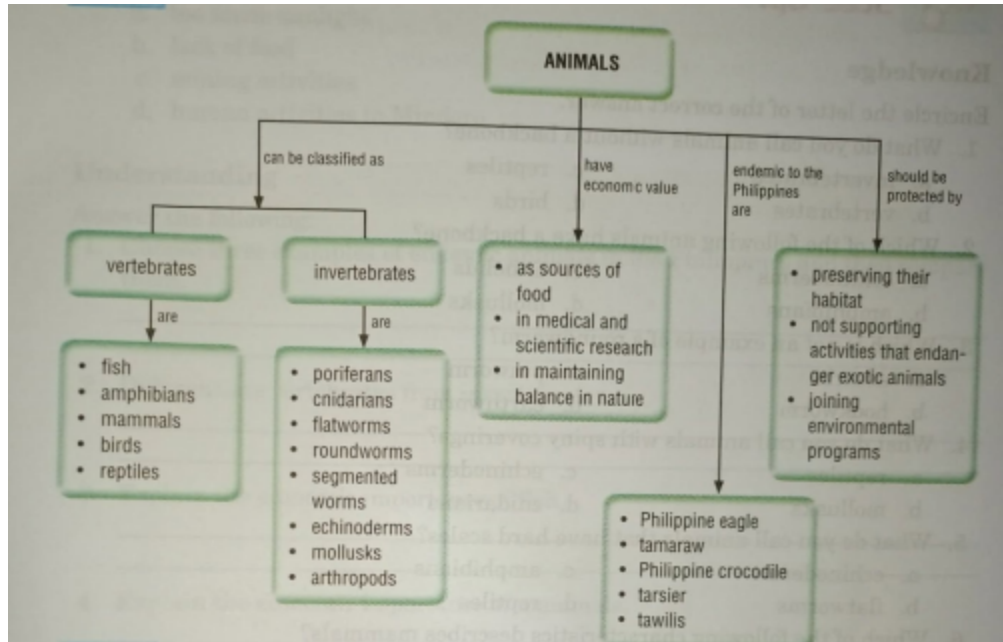
Answer the question below. Write your answer in the space provided. Do you support the use of live animals in experiments? Why?

How You Learned

How did you keep your interest while doing the activity?

Express Your Learning

List five (5) things that you can do to care for and protect animals.



Wrap Up

Animals are classified into vertebrates and invertebrates.

Vertebrates are animals with a backbone.

Invertebrates are animals without a backbone.

Vertebrates are classified into mammals, birds, reptiles, amphibians, and fish.

Invertebrates are classified into poriferans, cnidarians, flatworms, roundworms, segmented worms, echinoderms, mollusks, and arthropods.

Vertebrates can be sources of food and antivenoms and used for medical and environmental research. These animals can also help control pests and play a role in keeping the balance in nature.

Invertebrates can be sources of food. They can help maintain balance in nature and help in the nutrient cycle. Invertebrates may also serve as pollinators.

Some animals endemic in the Philippines include Philippine eagle, tamaraw, Philippine crocodile, tarsier, and tawilis.

Caring for and protecting animals can be expressed by preserving their habitats, not supporting activities that endanger exotic animals as pets, and joining tal programs. Mikaella

Size Up!

Knowledge

Encircle the letter of the correct answer.

1. What do you call animals without a backbone?

- a. invertebrates
- c. reptiles
- b. vertebrates

d. birds

2. Which of the following animals have a backbone?

a. echinoderms

c. annelids

b. amphibians

d. Mollusss

3. Which is not an example of a roundworm?

a. ascaris

c. pinworm

b. hookworm

d. Earthworm

4. What do you call animals with spiny coverings?

a reptiles

c. echinoderms

b. mollusks

d. Cnidarians

5. What do you call animals that

have hard scales?

a echinoderms

c. amphibians

b. flatworms

d. Reptiles

6. Which of the following characteristics describes mammals?

a. Mammals are covered with exoskeleton.

b. Mammals give birth to their young.

c. Mammals have a segmented body.

d. Mammals can pollinate flowers.

7. An example of a cnidarian is a jellyfish because

a. it is a vertebrate; in its life cycle it will undergo a medusa and polyp form

b. it has a body with segments or rings that are joined together

d. it is able to lay eggs

c. it is similar to other fishes of the sea

8. The process of metamorphosis can be found in arthropods

- a. mammals
- b. nematodes
- c. amphibians
- d. Porifera

9. Which of the following statements is true about fish?

- b. Fish have scales.
- a. Fish can produce milk.
- c. Fish can live both on land and in water.
- d. Fish can fly.

10. The tamaraw lives in the island of Mindoro. It is now under threat because of

- a. too much sunlight
- b. lack of food
- c. mining activities
- d. human activities in Mindoro

Understanding

Answer the following:

1. Choose three examples of endemic animals in the Philippines and then compare them.
2. Differentiate vertebrates from invertebrates.
3. Explain the economic importance of fish.
4. Explain the economic importance of mammals.
5. Cite at least two situations that show how people protect and care for animals.

Product

Create a slogan that would encourage others to treat animals with care and compassion. The slogan must be short but catchy and memorable. Write your slogan on a sheet of bond paper. Design and decorate your slogan to make it attractive and interesting.

